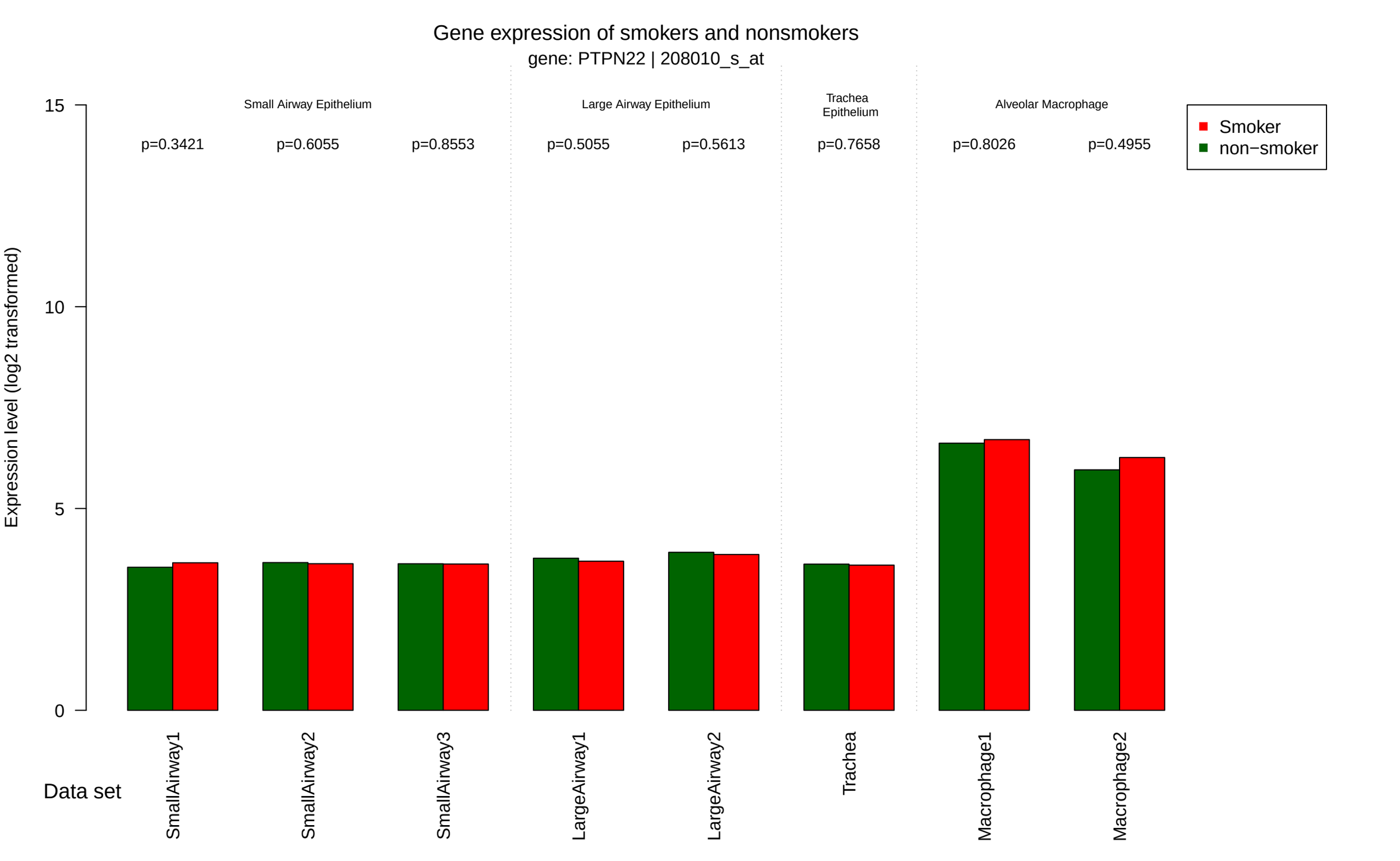
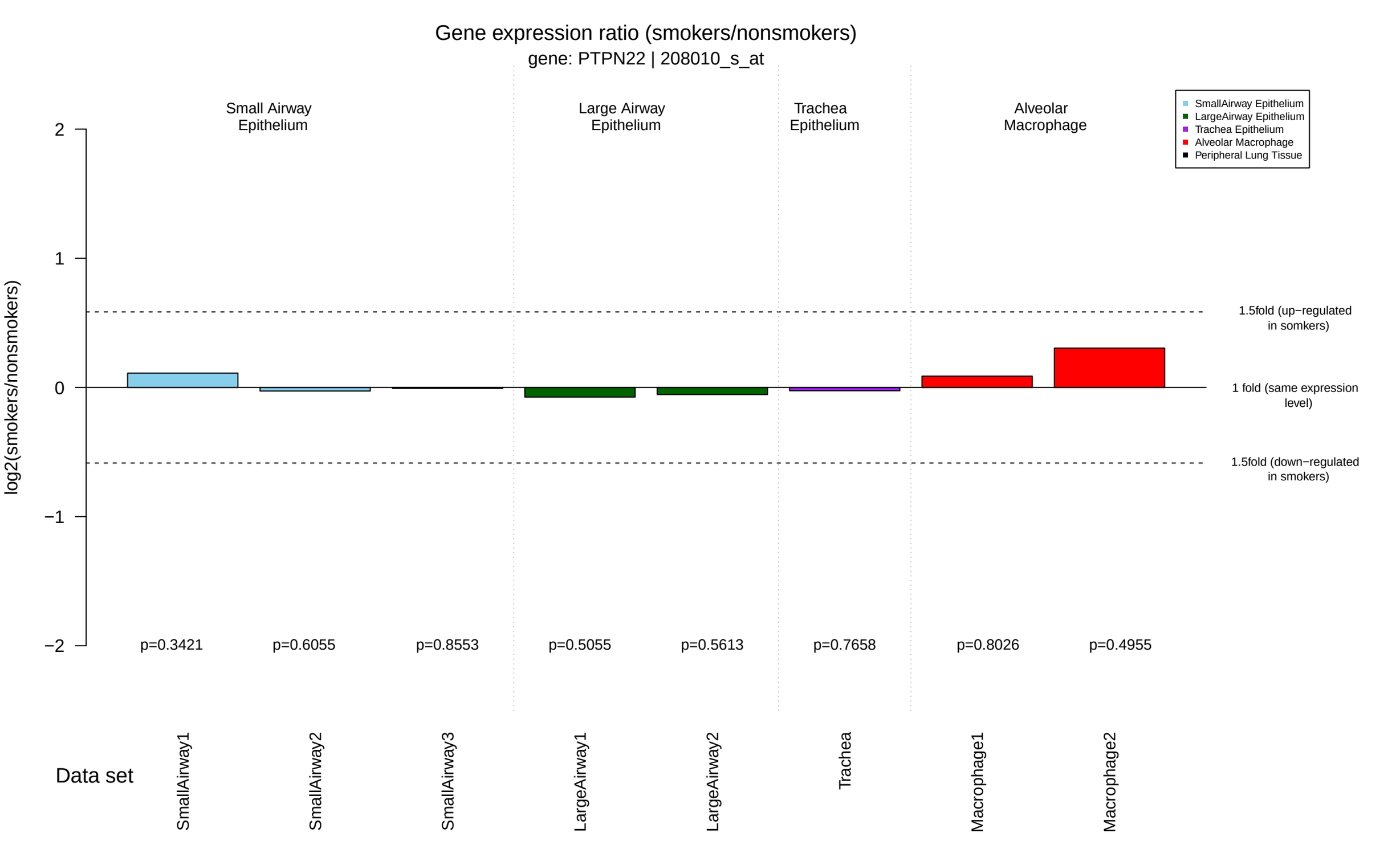
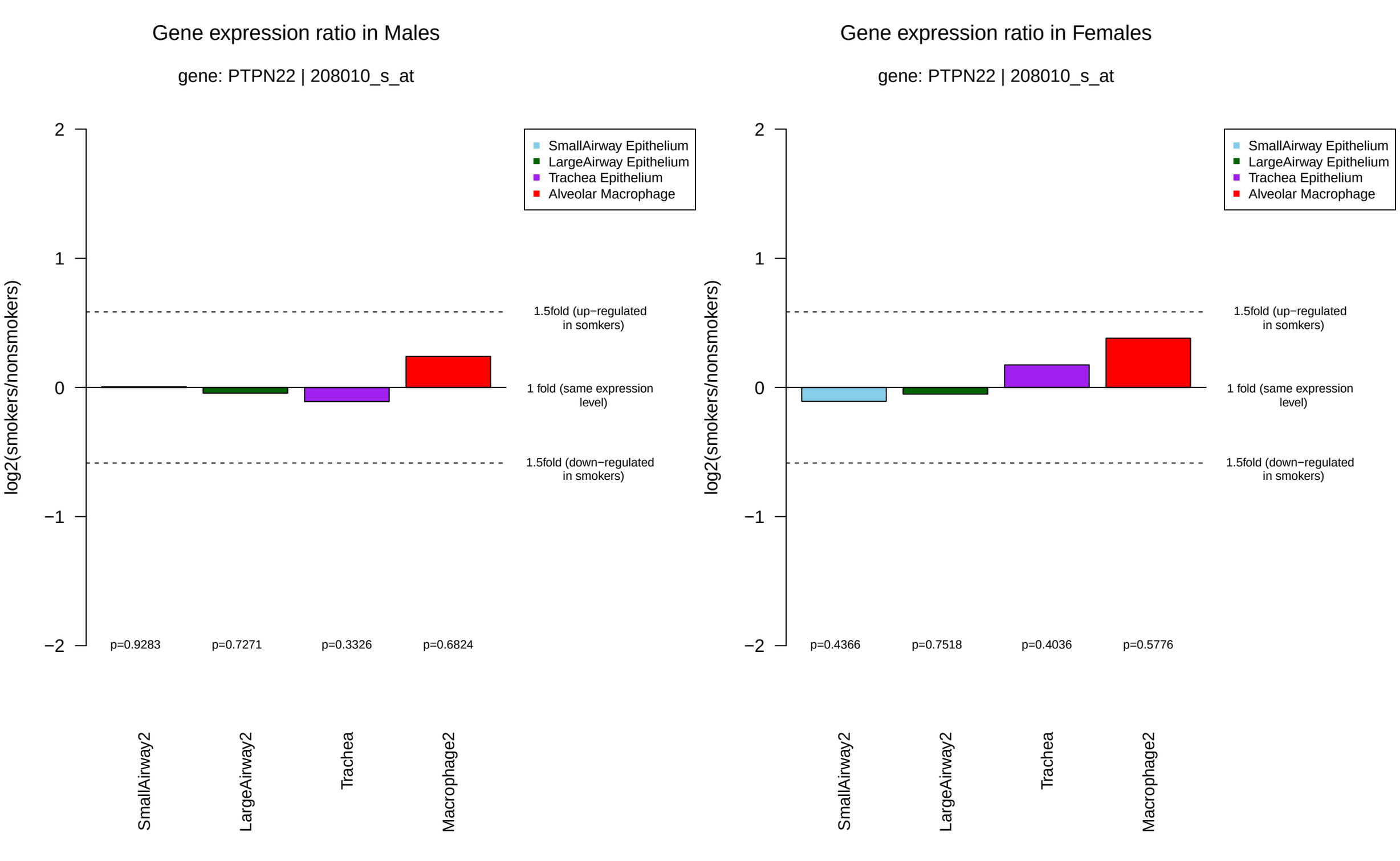
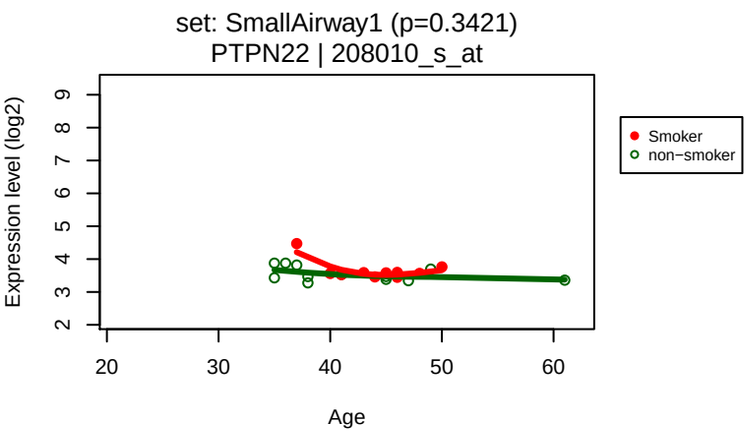
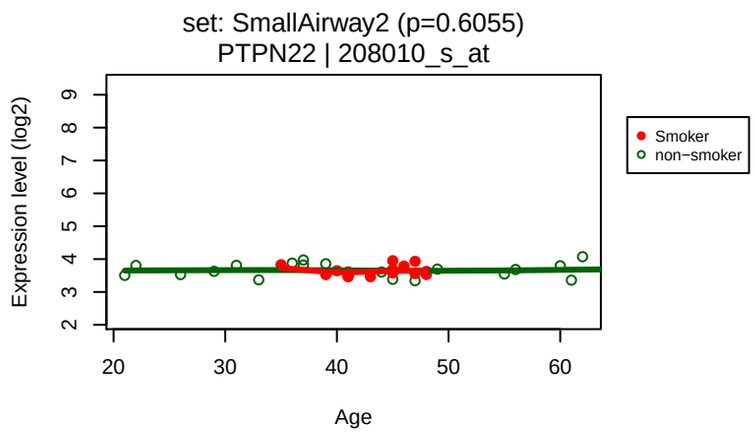
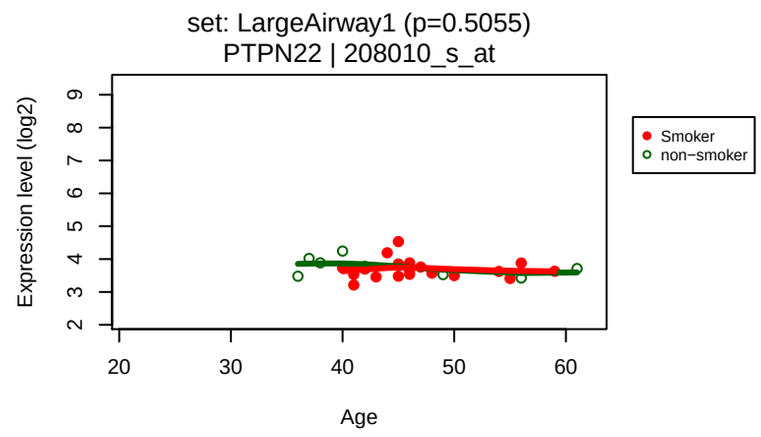

Gene expression of smokers and nonsmokers
gene: PTPN22 | 208010_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3421
p=0.6055
p=0.8553
p=0.5055
p=0.5613
p=0.7658
p=0.8026
p=0.4955
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

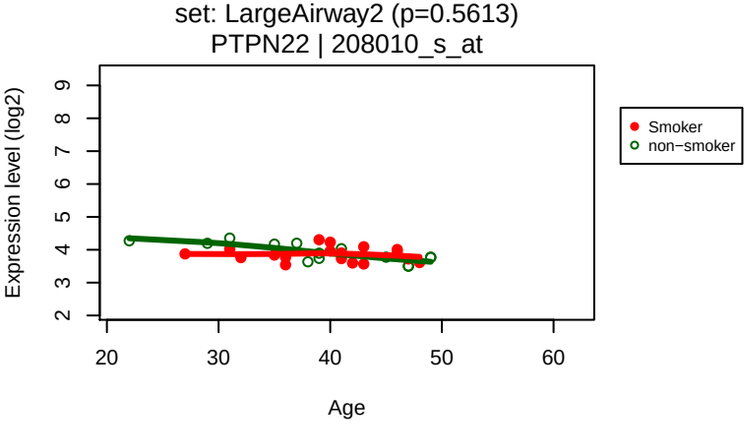

Gene expression ratio (smokers/nonsmokers)
gene: PTPN22 | 208010_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3421
p=0.6055
p=0.8553
p=0.5055
p=0.5613
p=0.7658
p=0.8026
p=0.4955
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

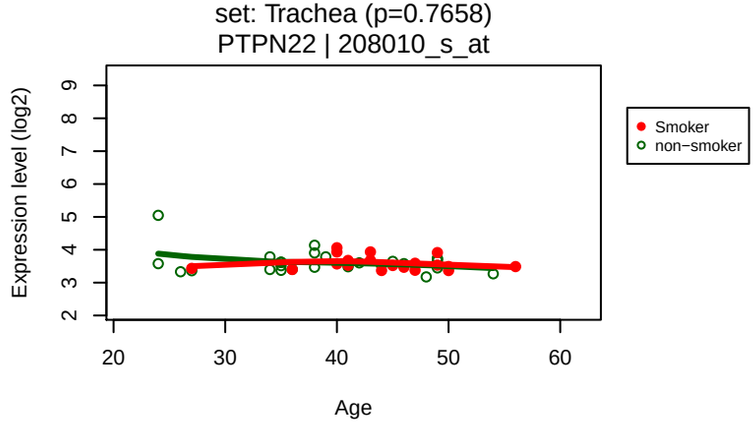

Gene expression ratio in Males
gene: PTPN22 | 208010_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9283
p=0.7271
p=0.3326
p=0.6824
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTPN22 | 208010_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4366
p=0.7518
p=0.4036
p=0.5776
SmallAirway2
LargeAirway2
Trachea
Macrophage2

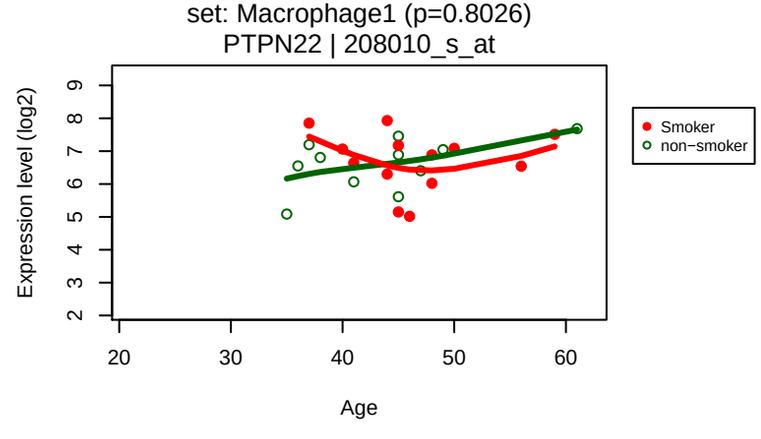

set: SmallAirway1 (p=0.3421)
PTPN22 | 208010_s_at
Expression level (log2)
Age
Smoker
non−smoker

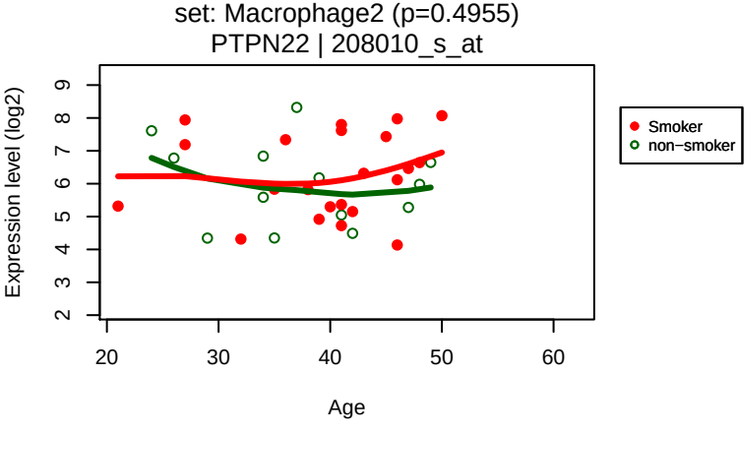

set: SmallAirway2 (p=0.6055)
PTPN22 | 208010_s_at
Expression level (log2)
Age
Smoker
non−smoker

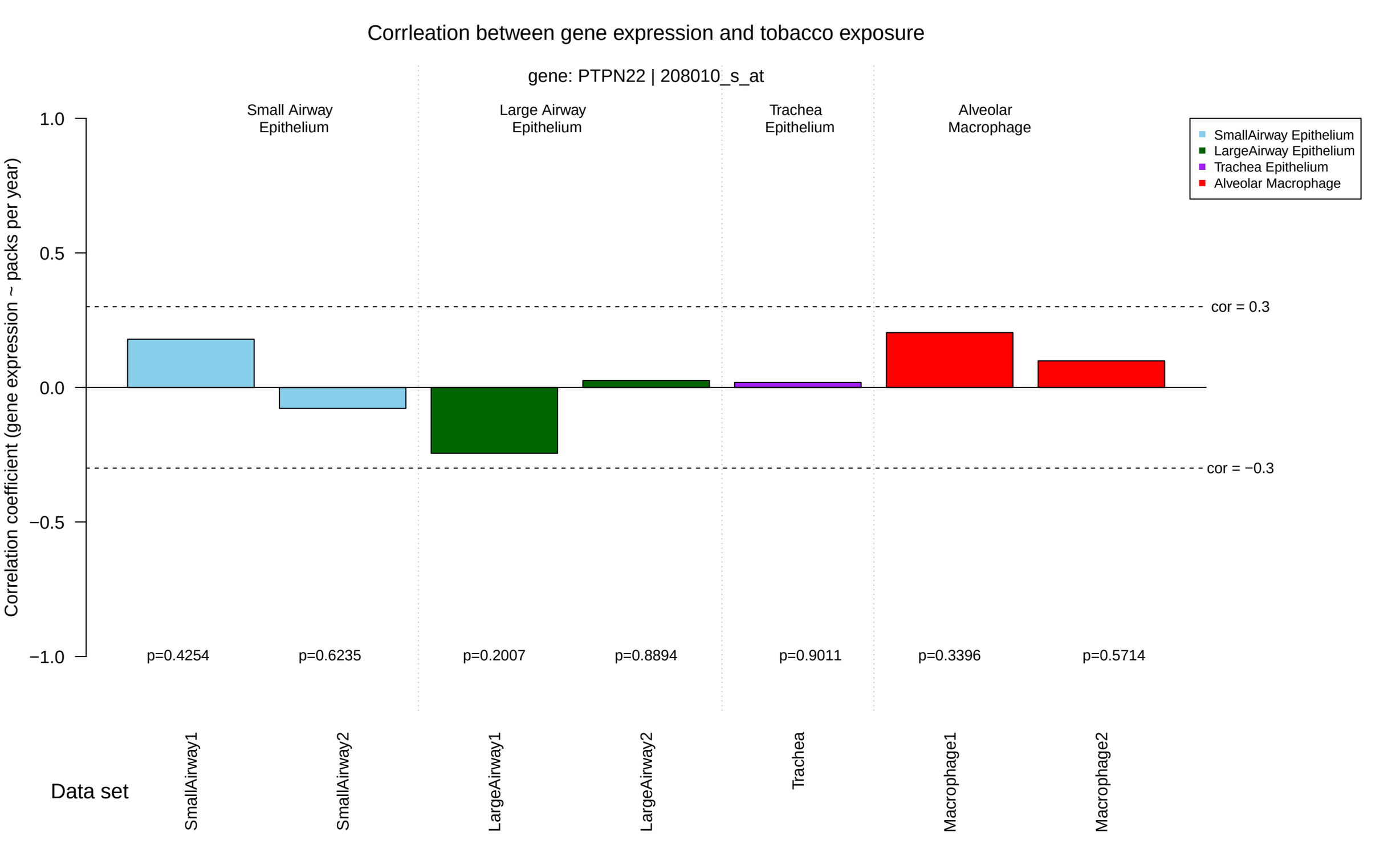

set: LargeAirway1 (p=0.5055)
PTPN22 | 208010_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5613)
PTPN22 | 208010_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7658)
PTPN22 | 208010_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8026)
PTPN22 | 208010_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4955)
PTPN22 | 208010_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTPN22 | 208010_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4254
p=0.6235
p=0.2007
p=0.8894
p=0.9011
p=0.3396
p=0.5714
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2